

RoadMap of PeoplePulse AI

AI-Powered Employee Retention & Workforce Intelligence Platform

Goal

Help HR teams answer questions like:

- Why are employees leaving?
 - Who is likely to resign?
 - Which teams are burning out?
 - Who deserves promotion?
 - Is salary fair?
 - Is training effective?
 - Which departments need hiring?
 - What should HR do next?
-

COMPLETE ROADMAP

Planning



Database Design



Dataset Collection



Excel



SQL



Python Analytics



Machine Learning



Power BI



Tableau



Streamlit Web App



Local AI Assistant



Flutter Mobile App



Deployment

PHASE 1 — PRODUCT DESIGN

This phase is where we'll define everything before coding.

Deliverables

- Software Requirements Specification (SRS)
 - Database ER Diagram
 - User Flow
 - UI Wireframes
 - Folder Structure
 - Color Palette
 - Logo
 - Tech Stack
 - GitHub Repository
-

PHASE 2 — DATABASE DESIGN

This is the heart of the project.

Instead of one CSV, we'll create a relational HR database.

Tables

Employee

- EmployeeID
- First Name
- Last Name
- Email
- Phone
- DOB
- Age
- Gender
- DepartmentID
- ManagerID
- Designation
- Join Date
- Salary
- Employment Type
- Experience
- Work Mode
- Status

Department

- DepartmentID
- Department Name
- Budget

- Head
 - Location
-

Attendance

- EmployeeID
 - Attendance %
 - Work Hours
 - Overtime
 - Late Arrivals
-

Performance

- EmployeeID
 - Performance Score
 - Projects Completed
 - KPI Score
 - Manager Rating
 - Peer Rating
 - Promotion Date
-

Salary History

- EmployeeID
 - Current Salary
 - Previous Salary
 - Increment
 - Bonus
-

Training

- EmployeeID
 - Course
 - Hours
 - Completion
 - Certification
-

Leave

- EmployeeID
 - Sick Leave
 - Casual Leave
 - Paid Leave
 - Unpaid Leave
-

Satisfaction Survey

- Job Satisfaction
 - Work Life Balance
 - Environment
 - Manager Support
 - Recognition
 - Career Growth
-

Attrition

- EmployeeID
 - Resigned
 - Exit Date
 - Exit Reason
-

Recruitment

- Candidate
- Interview Score
- Joining Date
- Recruiter

PHASE 3 — DATASET

We'll create a realistic company dataset.

Approximately

- 5,000 Employees
- 10 Departments
- 15 Managers
- 3 Years of History

It should look like company data—not Kaggle demo data.

PHASE 4 — EXCEL

Most students skip this.

We'll include:

- Cleaning
- Data Validation
- Conditional Formatting
- Pivot Tables
- Pivot Charts
- Dashboard
- Lookup Functions
- KPI Calculations
- Slicers

PHASE 5 — SQL

This will impress interviewers.

Database

MySQL

Topics

- Database
- Tables
- Primary Keys
- Foreign Keys
- Constraints
- Views
- Joins
- Group By
- CTEs
- Window Functions
- Stored Procedures
- Triggers

Business Queries

Not 10

Not 20

We'll write around **80–100 business queries**.

Examples

- Highest Attrition Department
- Lowest Salary Team
- Promotion Gap

- Employees Eligible for Promotion
 - Employees with Burnout
 - Training Effectiveness
 - Average Salary
 - Monthly Hiring
 - Gender Ratio
 - Leave Analysis
-

PHASE 6 — PYTHON

Libraries

Pandas

NumPy

Matplotlib

Plotly

Scikit-learn

Statsmodels

Data Cleaning

- Missing Values
 - Duplicate Records
 - Outliers
 - Encoding
 - Scaling
-

EDA

- Correlation
- Distribution

- Department Analysis
 - Salary Analysis
 - Satisfaction
 - Burnout
 - Attrition
 - Gender
 - Experience
-

Statistics

- Mean
 - Median
 - Mode
 - Variance
 - Hypothesis Testing
 - Correlation
-

Machine Learning

Attrition Prediction

Models

- Logistic Regression
 - Decision Tree
 - Random Forest
 - XGBoost (optional)
-

Burnout Prediction

Promotion Prediction

Salary Prediction

Hiring Forecast

Clustering

Employee Segmentation

- High Performers
 - Burnout Risk
 - Future Leaders
 - Loyal Employees
-

Forecasting

- Attrition
 - Hiring
 - Salary Budget
-

PHASE 7 — POWER BI

This should look enterprise-level.

Dashboard Pages

1

Executive Overview

2

Employee Analytics

3

Department Analytics

4

Attrition

5

Burnout

6

Performance

7

Training

8

Compensation

9

Forecast

10

Recommendations

Features

Bookmarks

Drill Through

Tooltips

Dynamic KPIs

Navigation

Dark Theme

PHASE 8 — TABLEAU

Not duplicate dashboards.

Instead

A Story.

Story Flow

Company

↓

Employees

↓

Attrition

↓

Burnout

↓

Recommendations

PHASE 9 — STREAMLIT WEB APP

This becomes the actual software.

Login

HR

Manager

CEO

Sidebar

Dashboard

Employees

Departments

Analytics

Predictions

Reports

AI Assistant

Settings

Employee Management

CRUD

- ☒ Add Employee
- ☒ Edit Employee
- ☒ Delete Employee
- ☒ Update Salary
- ☒ Assign Training
- ☒ Promote
- ☒ Add Leave

Analytics

Charts

KPIs

Predictions

Filters

Reports

Generate PDF

Download CSV

PHASE 10 — LOCAL AI

Using Ollama

Llama 3.2 or Gemma

The AI does **not** analyze raw data.

Python analyzes.

AI explains.

Example

Question

Why is Sales losing employees?

Python retrieves metrics.

AI explains in plain English.

PHASE 11 — MOBILE APP

Flutter

Home

KPIs

Notifications

Department Health

Employee Search

Reports

Alerts

Notifications

- Burnout Increased

- Promotion Due
- Attrition Increased
- Hiring Needed

PHASE 12 — DEPLOYMENT

GitHub

Streamlit Cloud / Render

Power BI Service

Portfolio Website

AI FEATURES

These are mandatory.

Workforce Health Score

Attrition Risk Score

Burnout Score

Promotion Readiness Score

Salary Fairness Score

Department Health Score

AI Recommendation Engine

Executive Summary Generator

Natural Language Assistant

ADDITIONAL FEATURES

Dark Mode

Export PDF

Export Excel

Download Reports

Advanced Filters

Role-Based Access

Search Employees

Sorting

Notifications

Responsive UI

Audit Logs

Activity Timeline

BONUS FEATURES

- Employee 360° Profile
- Manager Dashboard
- Organization Hierarchy
- Department Comparison
- Salary Simulator ("What if we increase salaries by 10%?")
- Attrition Cost Calculator
- Hiring Budget Estimator
- Workforce Diversity Dashboard
- Skills Matrix

- Succession Planning Dashboard

PROJECT STRUCTURE

PeoplePulse-AI/

|

|— docs/

|

|— SRS

|

|— ERD

|

|— Architecture

|

|— API Docs

|

|— dataset/

|

|— excel/

|

|— sql/

|

|— python/

|

|— cleaning

|

|— eda

|

|— ml

|

|— forecasting

|

|— recommendation_engine

|

|— powerbi/

|

|— tableau/

```
|
|— streamlit/
|
|— mobile/
|
|— reports/
|
|— assets/
|
|— README.md
```

What will make this project truly stand out

One thing I would add that isn't in the list above is a feature inspired by enterprise HR platforms:

Scenario Simulator

HR can ask questions like:

- "What happens if we increase Engineering salaries by 10%?"
- "What if we hire 20 more developers?"
- "What if we reduce overtime by 25%?"
- "What if we promote all employees waiting more than 3 years?"

Your Python models recalculate the projected attrition, burnout, and workforce health scores and show the expected impact.

This transforms your project from a reporting dashboard into a **decision-support system**, which is exactly the kind of capability that impresses interviewers because it demonstrates business thinking, not just technical skills.

How I suggest we work together

We'll build this exactly like a software company would:

- **Phase 1:** Product planning and UI/UX
- **Phase 2:** Database and dataset

- **Phase 3:** SQL and Excel
- **Phase 4:** Python analytics and ML
- **Phase 5:** Power BI and Tableau
- **Phase 6:** Streamlit application
- **Phase 7:** Local AI assistant
- **Phase 8:** Mobile app
- **Phase 9:** Testing, deployment, and interview preparation

I also recommend keeping a project journal as we build. By the end, you'll not only have the application but also the diagrams, documentation, GitHub commits, and explanations needed to confidently discuss every design decision during your internship interviews.